

Chapter - 4

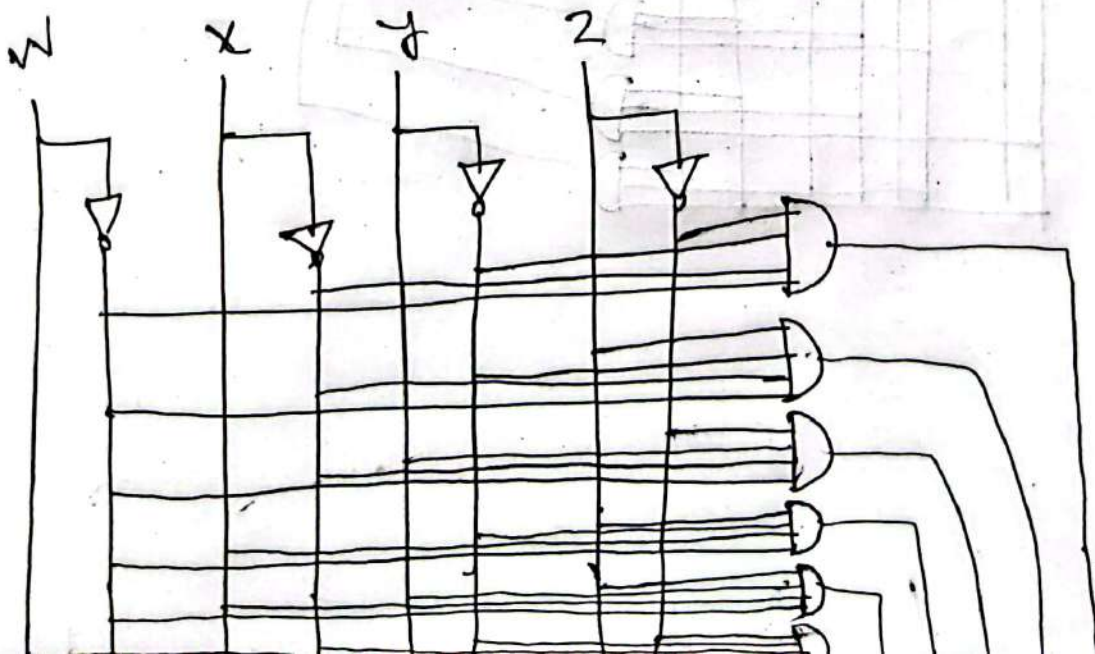
w	x	y	z	F
0	0	0	0	1
0	0	0	1	1
0	0	1	0	1
0	0	1	1	1
0	1	0	0	1
0	1	0	1	1
0	1	1	0	1
0	1	1	1	1
1	0	0	0	1
1	0	0	1	1
1	0	1	0	1
1	0	1	1	1
1	1	0	0	1
1	1	0	1	1
1	1	1	0	1
1	1	1	1	1

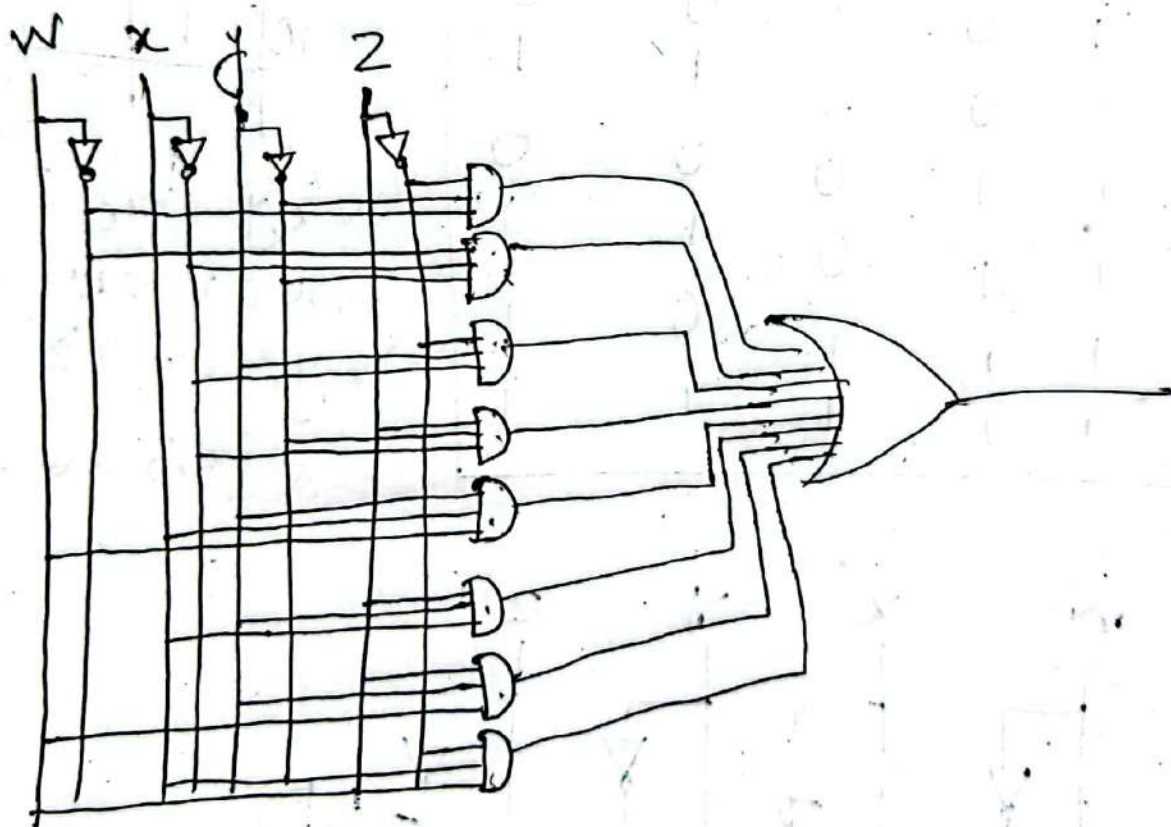
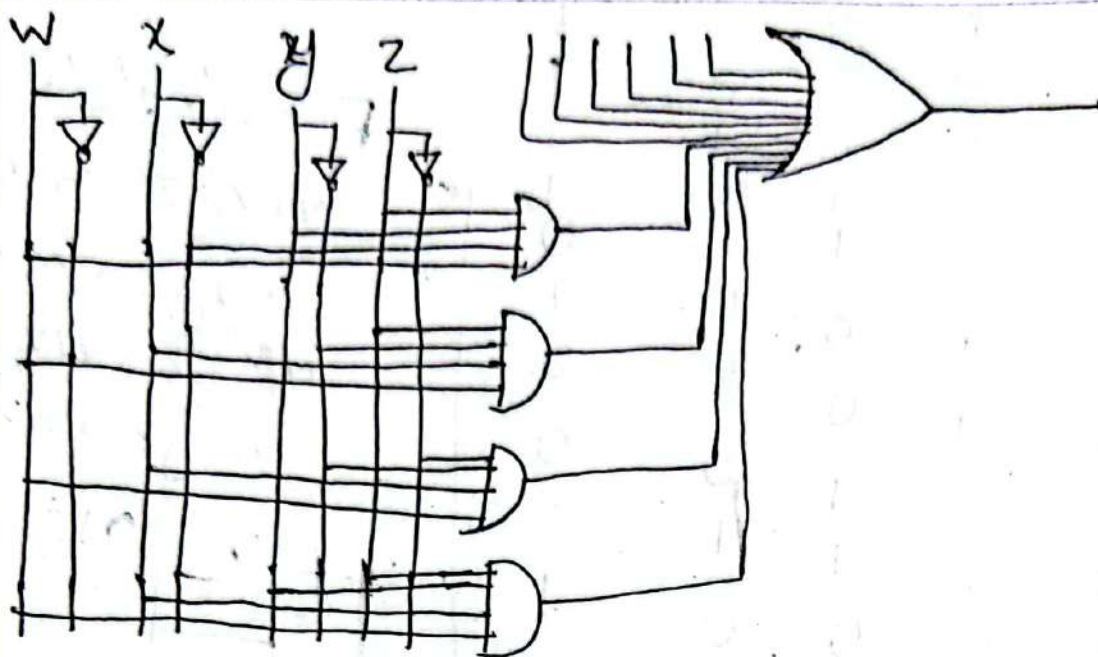
$$F = w'x'y'z' + w'x'y'z + w'x'yz' + w'x'yz + wx'y'z' + wx'y'z + wx'yz' + wx'yz + wxyz' + wxyz + w'xyz' + w'xyz + w'x'yz' + w'x'yz + wx'yz' + wx'yz + wxyz' + wxyz$$

	00	01	11	10
00	1	1		1
01	1		1	
11		1	1	1
10	1		1	

$$F = w'x'y'z' + w'x'y'z + w'x'yz' + w'x'yz + wx'y'z' + wx'y'z + wx'yz' + wx'yz + wxyz' + wxyz + w'xyz' + w'xyz + w'x'yz' + w'x'yz + wx'yz' + wx'yz + wxyz' + wxyz$$

$$= \Sigma (3, 5, 6, 9, 10, 12)$$





#2	x	y	z	D ₆	D ₅	D ₄	D ₃	D ₂	D ₁	D ₀
	0	0	0	0	0	0	0	0	0	0
	0	0	1	0	0	0	0	0	0	0
	0	1	0	0	0	0	1	0	0	0
	0	1	1	0	0	1	0	0	0	0
	1	0	0	0	0	0	0	0	0	0
	1	0	1	0	0	0	0	0	0	0
	1	1	0	0	0	0	0	0	0	0
	1	1	1	0	0	0	0	0	0	0

D₂

z	00	01	11	10
0		1	1	
1		1	1	

D₂

z	00	01	11	10
0				
1				

D₃

z	00	01	11	10
0				1
1				1

D₄

z	00	01	11	10
0			1	
1		1		

D₅

z	00	01	11	10
0				
1	1	1	1	

D₆

z	00	01	11	10
0				
1			1	1

3. a) required output : w, x, y, z

a_1	a_0	b_1	b_0	w	x	y	z
0	0	0	0	0	0	0	0
0	0	0	1	0	0	0	0
0	0	1	0	0	0	0	0
0	0	1	1	0	0	0	0
0	1	0	0	0	0	0	0
0	1	0	1	0	0	0	0
0	1	1	0	0	0	0	0
0	1	1	1	0	0	0	0
1	0	0	0	0	0	0	0
1	0	0	1	0	0	0	0
1	0	1	0	0	0	0	0
1	0	1	1	0	0	0	0
1	1	0	0	0	0	0	0
1	1	0	1	0	0	0	0
1	1	1	0	0	0	0	0
1	1	1	1	0	0	0	0

	00	01	11	10
00				
01				
11				1
10			1	1

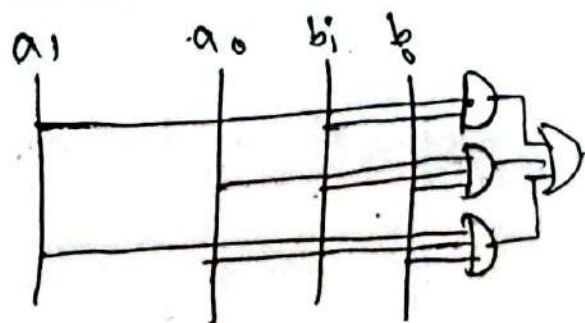
	00	01	11	10
00				
01			1	1
11		1		1
10		1	1	

$$y = a_1 b_0 b_1' + a_0 a_1' b_1 + a_0 b_1' b_1 + a_0' a_1 b_0$$

$$\therefore Z = a_0 b_0$$

$q_1 \backslash q_2$	00	01	11	10
00				
01			1	
11		1	1	1
10			1	1

$$\gamma = a_1 b_1 + a_0 b_1 b_0 + a_1 a_0 b_0$$



$a_1 \backslash b_1 b_0$	00	01	11	10
00			1	1
01		1		1
11	1		1	
10	1	1		

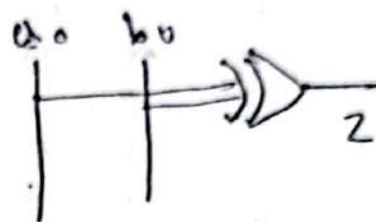
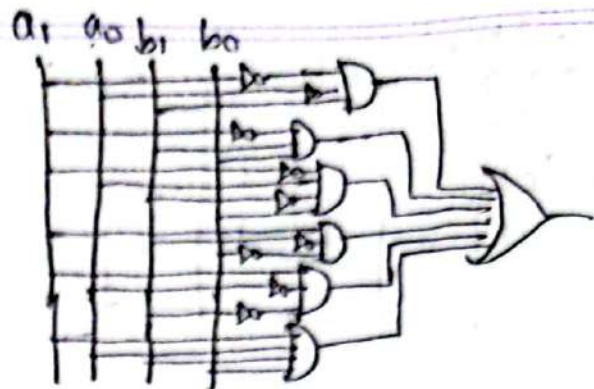
$$= a_1' a_0' b_1 + a_1' b_1 b_0 + a_1' a_0 b_1 b_0 + a_1 b_1 b_0' + a_1 a_0 b_1' + a_1 a_0 b_1 b_0$$

	00	01	11	10
00		1	1	
01	1			1
11	1			1
10		1	1	

$$Z = a_0' b_0 + a_0 b_0'$$

$$= a_0 \oplus b_0$$

5



Decimal

9's complement

5.

w	x	y	z	A	B	C	D
0	0	0	0	1	0	0	0
0	0	0	1	0	1	1	0
0	0	1	0	0	1	0	0
0	0	1	1	0	0	0	0
0	1	0	0	0	0	1	0
0	1	0	1	0	0	0	0
0	1	1	0	0	0	0	0
0	1	1	1	0	0	0	0
1	0	0	0	0	0	0	0
1	0	0	1	0	0	0	0

wxyz

w	x	y	z
00	01	11	10
00	1	1	
01			
11	x	x	x
10			x

$$A = w'x'y'$$

wxyz

w	x	y	z
00	01	11	10
00			1
01	1	1	
11	x	x	x
10			x

$$B = xy' + x'y$$

$$= x \oplus y$$

wxyz

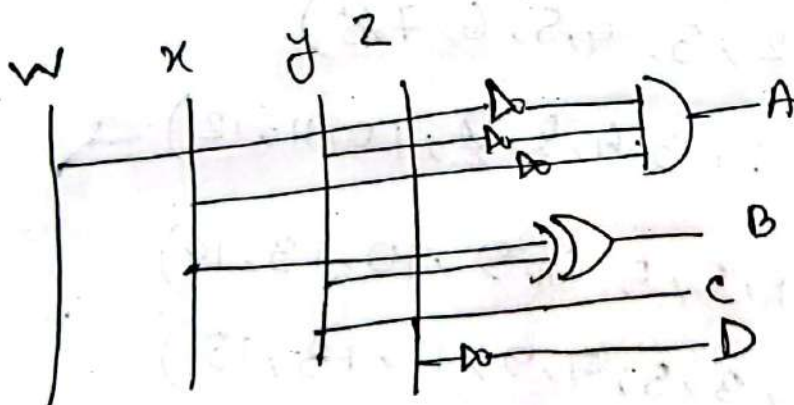
w	x	y	z
00	01	11	10
00			1
01			1
11	x	x	x
10			x

$$C = y$$

wxyz

w	x	y	z
00	01	11	10
00	1		
01			
11	x	x	x
10			x

$$D = z'$$



6. Binary number 2's complement

A	B	C	D	y_3	y_2	y_1	y_0
0	0	0	0	0	0	0	0
0	0	0	1	1	1	1	1
0	0	1	0	1	1	1	0
0	0	1	1	1	1	0	0
0	1	0	0	1	0	0	0
0	1	0	1	1	0	1	0
0	1	1	0	1	0	1	1
0	1	1	1	1	0	0	1
1	0	0	0	0	0	0	0
1	0	0	1	0	0	1	0
1	0	1	0	0	0	1	1
1	0	1	1	0	0	0	1
1	1	0	0	0	1	0	0
1	1	0	1	0	1	1	0
1	1	1	0	0	1	1	1
1	1	1	1	0	1	0	1

$$y_3 = \sum m(1, 2, 3, 4, 5, 6, 7, 8)$$

$$y_2 = \sum m(1, 2, 3, 4, 5, 9, 10, 11, 12)$$

$$y_1 = \sum m(1, 2, 5, 6, 9, 10, 13, 14)$$

$$y_0 = \sum m(1, 3, 5, 7, 9, 11, 13, 15)$$

$$y_3 = \bar{A}B + \bar{A}D + \bar{A}C + A\bar{B}\bar{C}\bar{D}$$

$$y_2 = \bar{B}C + \bar{B}D + B\bar{C}\bar{D}$$

$$y_1 = \bar{C}D + C\bar{D}$$

$$y_0 = D$$

7. Each digit of the decimal number is coded separately in binary form into 4 bit binary number. Since the circuit provides output that is 5 times the BCD digits. So, output value can contain maximum of 2 decimal digit or 8 binary digit. Let I_3-I_0 be the 4 bit BCD digit input to the $\times 5$ multiplier and H_3-H_0 and L_3-L_0 be the 8 bit BCD output.



Bcd input	Bcd output	J_3	J_2	J_1	J_0	H_3	H_2	H_1	H_0	L_3	L_2	L_1	L_0
0	00	0	0	0	0	0	0	0	0	0	0	0	0
1	05	0	0	0	1	0	0	0	0	0	1	0	1
2	10	0	0	1	0	0	0	0	1	0	0	0	0
3	15	0	0	1	1	0	0	0	1	0	1	0	0
4	20	0	1	0	0	0	0	1	0	0	0	0	1
5	25	0	1	0	1	0	0	1	0	0	1	0	0
6	30	0	1	1	0	0	0	1	1	0	1	0	1
7	35	0	1	1	1	0	0	1	1	0	1	0	0
8	40	1	0	0	0	0	1	0	0	0	0	1	0
9	45	1	0	0	1	0	1	0	0	0	0	1	1

K-map for H_3, L_3 and L_1

0	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0

$$H_3 = L_3 = L_1 = 0$$

Similarly we get: $H_2 = J_3$, $H_1 = J_2$,

$$H_0 = J_1, L_2 = L_0 = J_0$$

So, the whole function can be implemented without a single logic gate

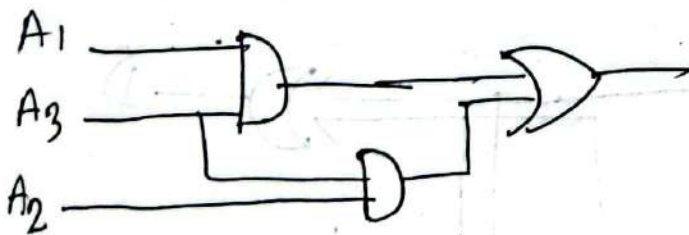
8. The truth table below contains all six entries for $F = 1$

A_3	A_2	A_1	A_0	F
1	0	1	0	1
1	1	0	0	1
1	1	0	1	1
1	1	1	0	1
1	1	1	1	1

Hence from the table

$$F = A_3(\bar{A}_2 A_1 \bar{A}_0 + \bar{A}_2 A_1 A_0 + A_2 \bar{A}_1 \bar{A}_0 + A_2 \bar{A}_1 A_0 + A_2 A_1 \bar{A}_0 + A_2 A_1 A_0)$$

$$F = A_3 A_2 + A_3 A_1$$



9. In full subtractor, Logical
for difference, $D = A \oplus B \oplus B_{in}$
Logical expression for:-

$$B_{out} = \bar{A}B + \bar{A}B_{in} + BB_{in}$$

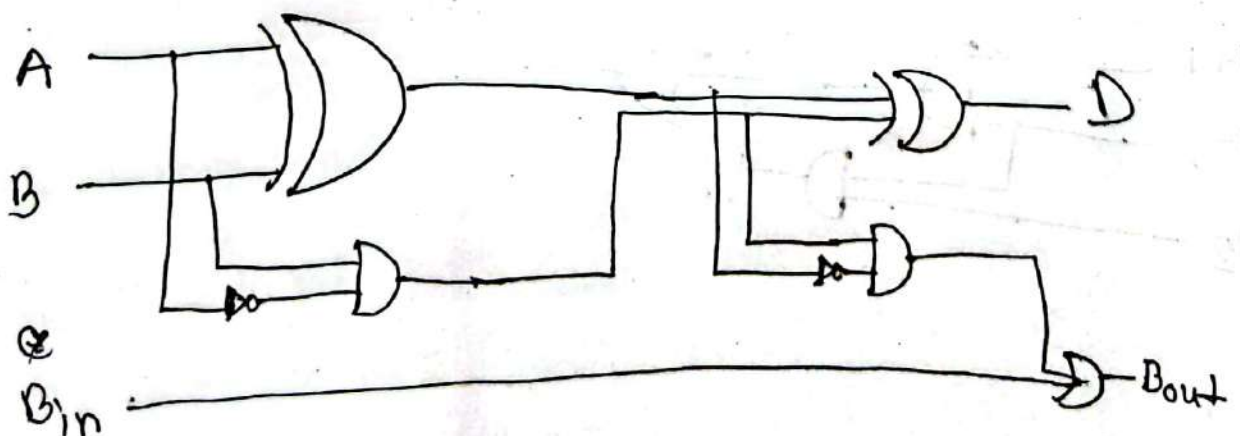
$$= \bar{A}B + \bar{A}B_{in}(B + \bar{B}) + BB_{in}(A + \bar{A})$$

$$= \bar{A}B + \bar{A}B B_{in} + \bar{A}\bar{B}\bar{B}_{in} + AB B_{in} + \bar{A}B B_{in}$$

$$= \bar{A}B + B_{in}(\bar{A}\bar{B} + AB)$$

$$= \bar{A}B + B_{in}(\overline{A \oplus B})$$

Logical circuit:



10	A	B	C	D	a	b	c	d	e	f	g
	0	0	0	0	0	-	-	0	0	-	0
	0	0	0	0	0	-	0	0	0	0	-
	0	0	1	0	0	-	-	0	0	0	-
	0	0	0	0	0	-	-	0	0	0	-
	0	1	0	0	0	0	-	0	0	0	-
	0	1	0	0	0	0	-	0	0	0	-
	0	1	1	0	0	0	-	0	0	0	-
	0	1	1	0	0	0	-	0	0	0	-
	1	0	0	0	1	-	-	0	0	0	-
	1	0	0	1	1	-	-	0	0	0	-

Full Adder

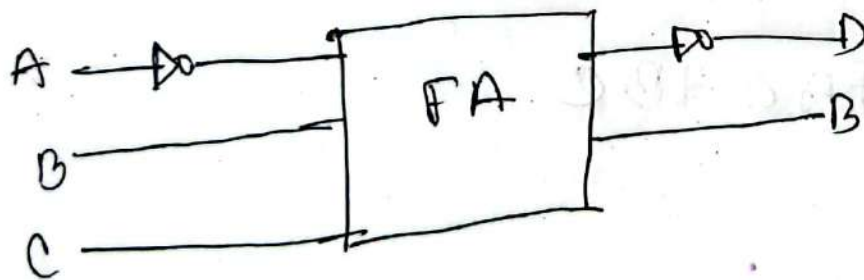
$$\text{sum} = A \oplus B \oplus C$$

$$\text{carry} = AB + BC + AC$$

Full subtractor

$$\text{Difference} = A \oplus B \oplus C$$

$$\text{Borrow} = \bar{A}B + BC + \bar{A}C$$



11: Herre Boolean

$$F_1 = \overline{(AB + BC + AC)} (A + B + C) + ABC$$

$$= (\overline{AB} \quad \overline{BC} \quad \overline{AC}) (A + B + C) + ABC$$

$$= (\overline{A} + B) \cdot (\overline{B} + C) (\overline{A} + C) (A + B + C) + ABC$$

$$= (\overline{A}\overline{B}) (\overline{A}\overline{C} + \overline{B}\overline{C} + \overline{B}\overline{C}) (\overline{A} + C) (A + B + C) + ABC$$

$$= \{ \overline{B}(1 + \overline{A}) + \overline{A}\overline{C} \} (\overline{A} + C) (A + B + C) + ABC$$

$$= \overline{A}\overline{B}C + \overline{A}B\overline{C} + A\overline{B}\overline{C} + ABC$$

$$F_2 = AB + AC + BC$$

	00	01	11	10	00	01	11	10
00	0	0	0	0	0	0	0	0
01	0	1	1	0	0	0	0	0
11	0	1	0	0	0	0	1	0
10	0	1	0	0	0	0	1	0
00	1	0	1	1	0	1	1	0
01	1	0	1	0	0	1	0	1
11	1	0	0	1	0	0	0	1
10	1	1	1	1	1	0	0	1

	00	01	11	10
00				
01				
11	x	x	1	x
10	1			

$$W = AB + AC'D$$

	00	01	11	10
00		x	x	x
01		1		1
11	x	x		x
10		1		1

$$Y = c'D + cD$$

$$= c \oplus D$$

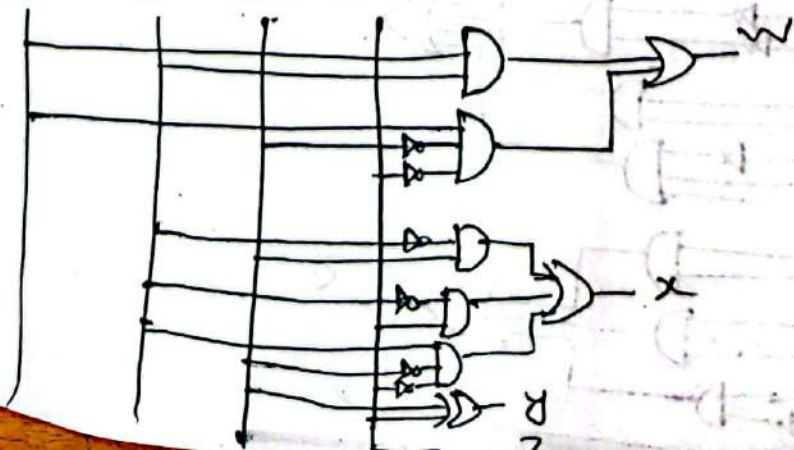
A B C D

	00	01	11	10
00		x	x	x
01	1			
11	x	x		x
10		1	1	1

$$X = B'C + B'D + BC'D'$$

	00	01	11	10
00		x	x	x
01		1	1	
11	x	x	x	x
10		1	1	

$$Z = D$$



12.

	5	2	4	2	1	8	4	-	2	-1
		A	B	C	D	W	X		Y	Z
0		0	0	0	0	0	0		0	0
1		0	0	0	1	0	1		1	1
2		0	0	1	0	0	1		1	0
3		0	0	1	1	0	1		0	1
4		0	1	0	0	0	1		0	0
5		0	1	0	1	1	0		1	1
6		1	1	0	0	1	0		1	0
7		0	1	1	1	1	0		0	1
8		1	1	1	0	1	0		0	1
9		1	1	1	1	1	1		1	1

AB \ CD	00	01	11	10
00				
01	x	1	1	x
11	x	x	1	1
10	x	x	x	x

	00	01	11	10
00	0	1	0	1
01	0	1	0	x
11	1	x	1	0
10	x	x	x	x

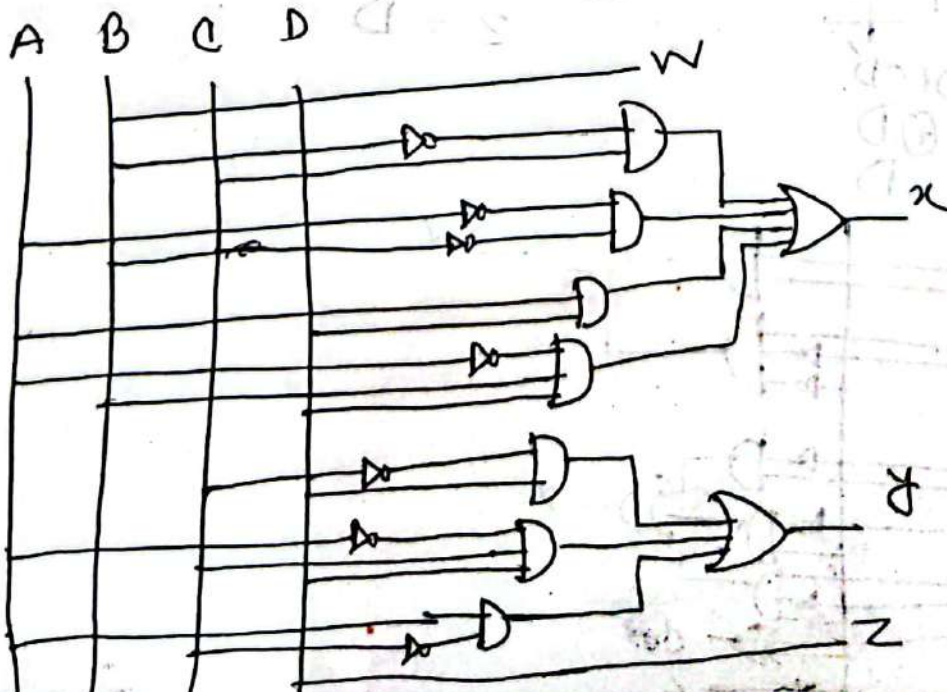
	00	01	11	10
00		1	1	
01		1	1	
11		x	1	
10	x	x	x	x

$Z = D$

$$Y = CD + A'CD' + AC'$$

	00	01	11	10
00		1	1	1
01	1			x
11		x	1	
10	x	x	x	x

$$X = B'C + A'B'D + AD + A'BD$$



14.

w	x	y	z	a	b	c	d	e	f	g
0	0	0	0	1	1	1	1	1	1	1
0	0	0	1	0	1	1	0	0	0	0
0	0	1	0	1	1	0	1	0	0	1
0	0	1	1	0	1	1	0	0	1	0
0	1	0	0	1	0	1	1	0	1	1
0	1	0	1	0	0	1	1	1	0	1
0	1	1	0	1	1	1	0	1	1	1
0	1	1	1	0	1	1	1	0	1	1
1	0	0	0	1	1	1	1	0	1	1
1	0	0	1	1	1	1	1	1	1	1

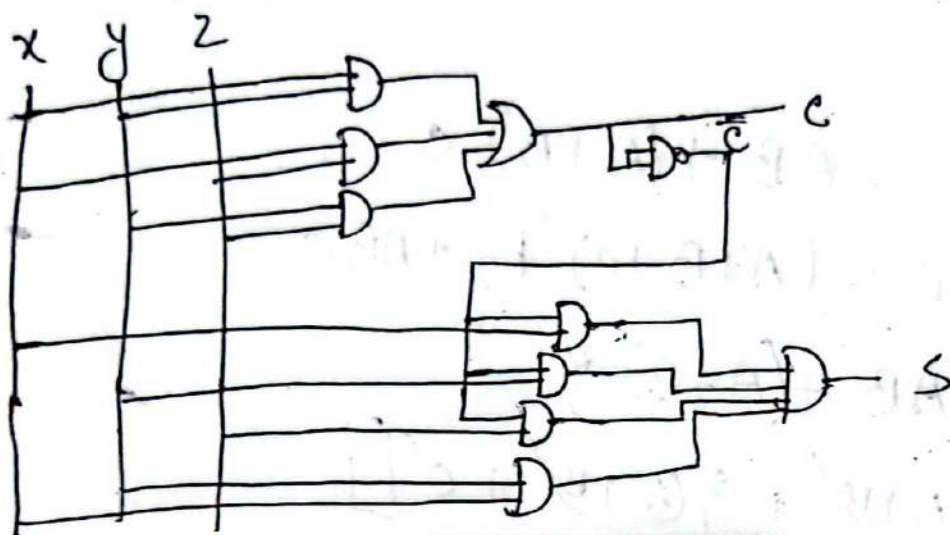
	00	01	11	10
00	1		1	1
01		1	1	1
11				
10	1	1		

	00	01	11	10
00	1	1	1	1
01	1		1	
11				
10	1	1		

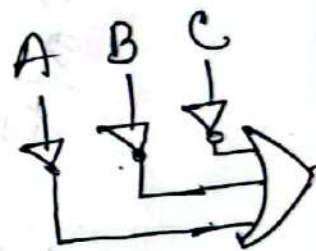
	00	01	11	10
00				1
01				
11	x	x	x	x
10			x	x

$$15. S = \bar{C}(x+y+z) + xyz$$

$$C = xy + xz + yz$$



$$\begin{aligned}
 16. F &= \overline{\bar{A} + \bar{B} + \bar{C}} + \overline{A + \bar{B} + \bar{B}} \\
 &= \overline{\bar{A} + \bar{B} + \bar{C}} + \overline{A + \bar{B} + \bar{B}} \\
 &= \bar{A} \cdot \bar{B} \cdot \bar{C} + \bar{A} \cdot \bar{B} + \bar{B} \\
 &= ABC\bar{C} + \bar{A}B + \bar{B} \\
 &= ABC\bar{C} + \bar{A} + \bar{B} [A + \bar{A}B = A + B] \\
 &= \bar{A} + \bar{B} + \bar{C}
 \end{aligned}$$



$$\begin{aligned}
 17. F_1 &= \overline{x \cdot xz \cdot y \cdot yz \cdot z \cdot zx} \\
 F_2 &= \overline{xy \cdot xz}
 \end{aligned}$$

18.

Decimal	BCD				Output			
	A	B	C	D	W	X	Y	Z

0
1
2
3
4
5
6
7
8
9

$$F_2 = AB + (A+B)C$$

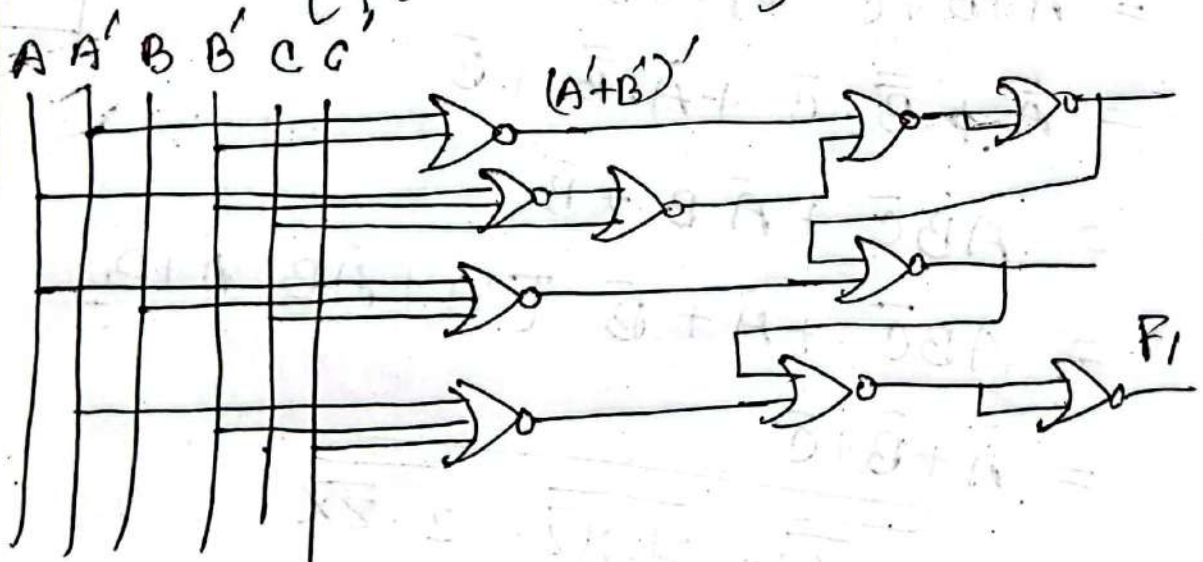
$$F_1 = (A+B+C)F_2 + ABC$$

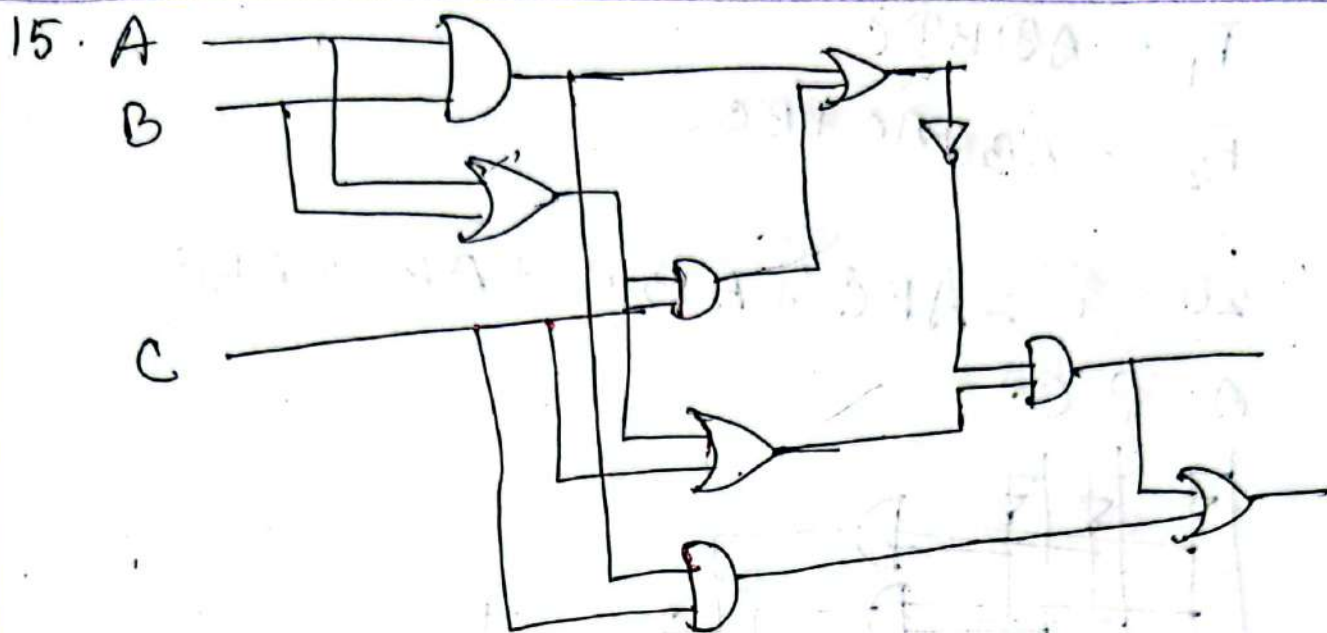
$$F_2 = \{AB + (A+B)C\}$$

$$= (A+B)' + \{ (A+B)' + C \}$$

$$F_1 = \{ (A+B+C)F_2' + ABC \}'$$

$$= \{ \{ (A+B+C)' + F_2 \}' + (A'+B'+C)' \}'$$





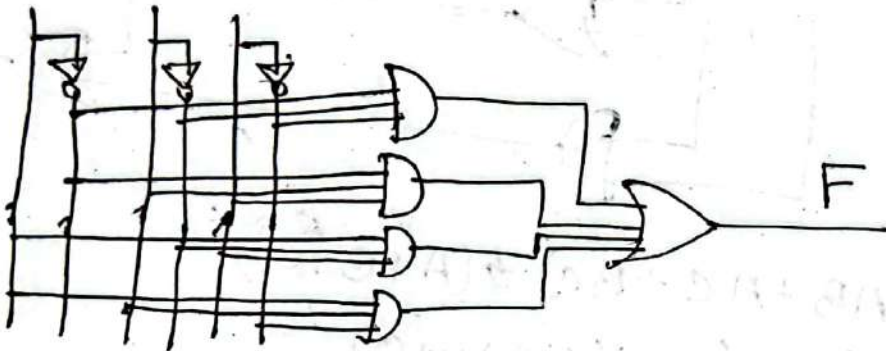
$$\begin{aligned}
 F_1 &= ABC + (AB + AC + BC)' (A + B + C) \\
 &= ABC + (AB)' (AC)' (BC)' (A + B + C) \\
 &= ABC + (A' + B') (A' + C') (B' + C') (A + B + C) \\
 &= ABC + (A' + B' + A'C' + B'B' + B'C') (A' + C') (A + B + C) \\
 &= \{ \bar{A}\bar{B} + \bar{A}\bar{C} + \bar{B}(1 + \bar{C}) \} (\bar{A} + \bar{C}) (A + B + C) + ABC \\
 &= (\bar{A}\bar{B} + \bar{B}\bar{C} + \bar{A}\bar{C}) (A + B + C) + ABC \\
 &= \bar{A}'\bar{B}'A + \bar{A}\bar{B}B + \bar{A}\bar{B}C + \bar{B}\bar{C}A + \bar{B}\bar{C}B + \bar{B}\bar{C}C + \bar{A}\bar{C}A + \\
 &\quad \bar{A}\bar{C}B + \bar{A}\bar{C}C + ABC \\
 &= \bar{A}\bar{B}C + \bar{A}B\bar{C} + B\bar{C} + \bar{A}\bar{C}B + ABC \\
 &= \bar{A}\bar{B}C + B\bar{C}(A + 1) + \bar{A}\bar{C}B + ABC \\
 &= \bar{A}\bar{B}C + A\bar{B}\bar{C} + \bar{A}\bar{C}B + ABC \\
 &= \bar{A}(\bar{B}C + B\bar{C}) + A(B\bar{C} + \bar{B}C) \\
 &= \bar{A}(B \oplus C) + A(B \oplus B)'
 \end{aligned}$$

$$F_1 = A \oplus B \oplus C$$

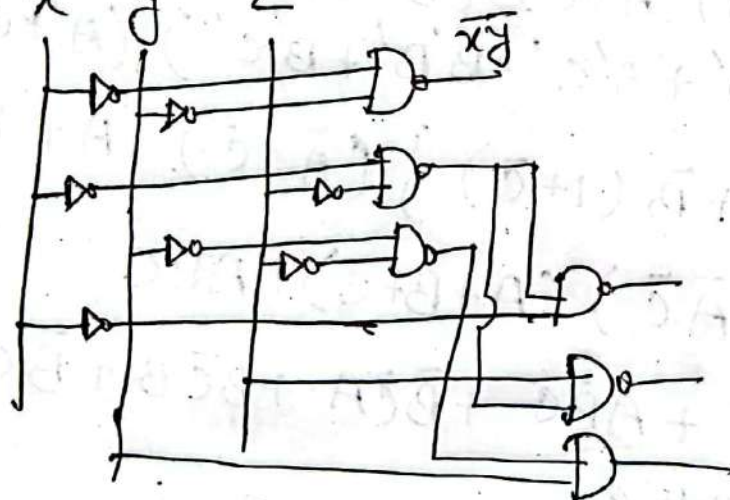
$$F_2 = AB + AC + BC$$

$$20. F = \bar{A}\bar{B}\bar{C} + \bar{A}B\bar{C} + A\bar{B}\bar{C} + ABC$$

A B C



21. x y z



$$\textcircled{1} \bar{x}y \quad \textcircled{2} \bar{x}z \quad \textcircled{3} \bar{y}z \quad \textcircled{4} \overline{\bar{x}z}$$

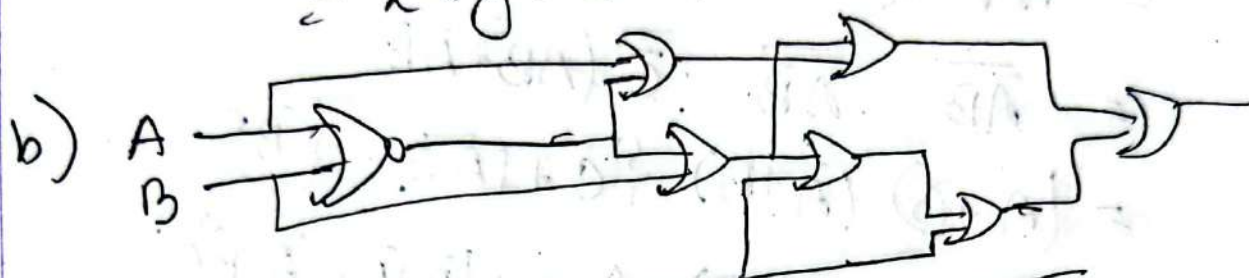
$$= \bar{x} \cdot \overline{\bar{x}z}$$

$$= \bar{x} + \overline{\bar{x}z}$$

$$\textcircled{5} \overline{\bar{x}z} = \overline{\bar{z} + xz} = \bar{z} + xz = \bar{z} + xz$$

$$\textcircled{6} \bar{y} = \overline{(yz)} = \bar{y}z + \bar{y}$$

$$\begin{aligned}
 F_1 &= \overline{4 \cdot 5 \cdot 6} \\
 &= \overline{\overline{x} \cdot \overline{xz}} + \overline{z \cdot \overline{xz}} + \overline{y \cdot \overline{yz}} \\
 &= \overline{\overline{x}} + \overline{\overline{xz}} + \overline{z} + \overline{\overline{xz}} + \overline{y} + \overline{\overline{yz}} \\
 &= x + y + z + \overline{xz} + \overline{xz} + \overline{yz}
 \end{aligned}$$



① $\overline{A+B}$ ② $\overline{A+A+B}$ 3. $\overline{B+A+B}$

$$= \overline{B} + (\overline{A+B})$$

$$= \overline{B} \cdot (A+B)$$

$$\begin{aligned}
 4 \text{ } \overline{C+3+2} &= \overline{\overline{C} + \overline{A} (A+B + \overline{B} \cdot (A+B))} \\
 &= \overline{C + A\overline{A} + \overline{A}B + A\overline{B} + B\overline{B}}
 \end{aligned}$$

$$= \overline{A\overline{B} + \overline{A}B} \cdot \overline{C}$$

$$= (\overline{A+B}) (A+\overline{B}) \cdot \overline{C}$$

$$= (\overline{A}\overline{B} + AB) \cdot \overline{C}$$

$$= \overline{A \oplus B} \cdot \overline{C}$$

$$5. \overline{2+3+4}$$

$$= \overline{\bar{A}(A+B) + \bar{B}(A+B) + \bar{C}(A+\bar{A}\bar{B})}$$

$$= \overline{A\bar{A} + \bar{A}B + A\bar{B} + B\bar{B} + \bar{C}(AB + \bar{A}\bar{B})}$$

$$= \overline{\bar{A}B + A\bar{B} + \bar{C}(AB + \bar{A}\bar{B})}$$

$$= \overline{\bar{A}B} \cdot \overline{A\bar{B}} \cdot \overline{\bar{C}(AB + \bar{A}\bar{B})}$$

$$= (A+B)(\bar{A}+\bar{B}) \{ \bar{C} + (\overline{\bar{A}B} \cdot \overline{A\bar{B}}) \}$$

$$= (A+B)(\bar{A}+\bar{B}) \{ \bar{C} + (\bar{A}+\bar{B})(A+B) \}$$

$$= (AB + \bar{A}\bar{B})(\bar{C} + \bar{A}B + A\bar{B})$$

$$= ABC + \bar{A}\bar{B}C$$

$$= C(AB + \bar{A}\bar{B})$$

$$= C(A \oplus B)$$

$$6. = 4+C$$

$$= \overline{\bar{C}(AB + \bar{A}\bar{B}) + C}$$

$$= \overline{AB\bar{C} + \bar{A}\bar{B}\bar{C} + C}$$

$$= \overline{AB\bar{C}} \cdot \overline{\bar{A}\bar{B}\bar{C}} \cdot \bar{C}$$

$$= (\bar{A}+\bar{B}+C)(A+B+C) \cdot \bar{C}$$

$$= (C + \bar{A}B + \bar{A}C + A\bar{B} + B\bar{C})\bar{C}$$

$$= \bar{A}B\bar{C} + A\bar{B}\bar{C}$$

$$= \bar{C}(A \oplus B)$$

$$F = \overline{B+B}$$

$$= \overline{C(\bar{A \oplus B}) + (A \oplus B)}$$

$$= \overline{C(\bar{A \oplus B})} \cdot \overline{C(A \oplus B)}$$

$$= \bar{C} + \overline{A \oplus B} \cdot \bar{C} + \overline{A \oplus B}$$

$$= \bar{C} + (\bar{A}B + A\bar{B})\bar{C} + \bar{A}B + A\bar{B}$$

$$= \bar{C} + (\bar{A}B + A\bar{B})\bar{C} + (A+B)(\bar{A}+B)$$